



ENDORSE

THE EUROPEAN DATA CONFERENCE ON REFERENCE DATA AND SEMANTICS



Visualisation and navigations of LOD graphs

- OP publishes public administration data
 - Web sites, PDFs
 - But also structured data using semantic and Link Open Data (LOD) principles
- LOD is difficult to navigate directly :
 - complexity of SPARQL syntax
 - complexity of data models/ontologies
 - no prior knowledge of what is “important” and “insignificant”
- Opportunity
 - Using visualisation

Visualisation and navigations of LOD graphs

■ Vinalod Project

- Preparatory Action - Link Open Data (PALOD)
- Contract using the “expert list” in Visualisation and web semantic.
- During years 2021-2023

• Scope

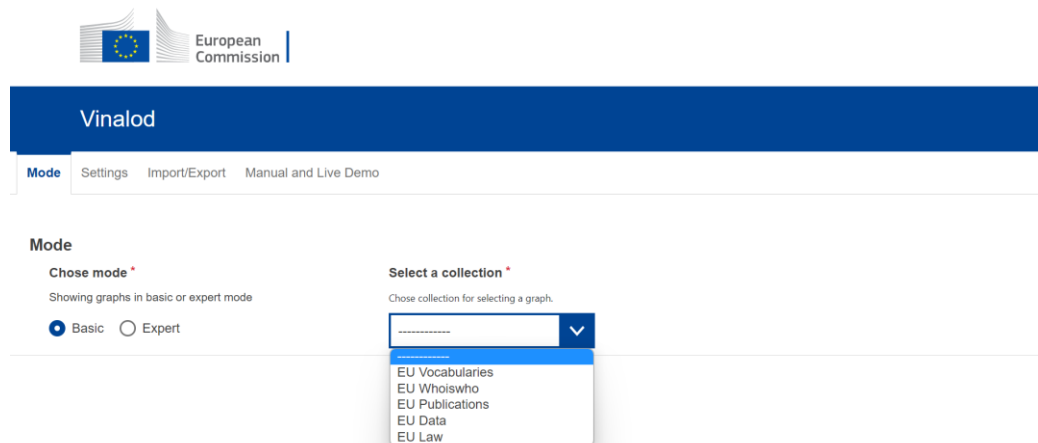
- All semantic data managed by Publications Office (Open data, reference vocabularies, EU Publications, EU Law, EU Whoiswho)
- Links to external triple stores like Wikidata

Visualisation and navigations of LOD graphs

- What output for the project?
 - Experimental visual tool to navigate LOD data
 - User-friendly/Intuitive (e.g. using icons and logos)
 - For citizen (basic mode) and also for experts (expert mode)
 - For OP data but also for external world data (e.g. Wikidata/DBpedia)
- The tool
 - Stand alone application that run on your PC/mobile
 - Dependent of SPARQL endpoints (CELLAR/ DEU and Wikidata)
 - Parametrizable:
 - new icons/logos to any URIs
 - new SPARQL queries
 - specific visualisations (e.g. tables / graphs)

Visualisation and navigations of LOD graphs

- Demonstration: Starting basic mode



The screenshot displays the Vinalod web application interface. At the top, the European Commission logo is visible. Below it, a dark blue header bar contains the text 'Vinalod'. A navigation bar below the header includes links for 'Mode', 'Settings', 'Import/Export', and 'Manual and Live Demo'. The 'Mode' section is active, showing two radio buttons: 'Basic' (selected) and 'Expert'. To the right, the 'Select a collection' dropdown menu is open, displaying a list of collections: 'EU Vocabularies', 'EU Whoiswho', 'EU Publications', 'EU Data', and 'EU Law'.

European Commission

Vinalod

Mode Settings Import/Export Manual and Live Demo

Mode

Chose mode *

Showing graphs in basic or expert mode

☒ Basic ☐ Expert

Select a collection *

Chose collection for selecting a graph.

EU Vocabularies
EU Whoiswho
EU Publications
EU Data
EU Law

Visualisation and navigations of LOD graphs

- Demonstration: EU Vocabularies graphs

Mode

Chose mode *

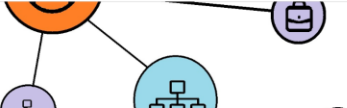
Showing graphs in basic or expert mode

☒ Basic ☐ Expert

Select a collection *

Chose collection for selecting a graph.

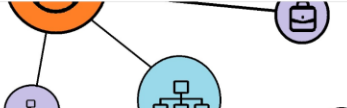
EU Vocabularies



Digital Europa Thesaurus

Digital Europa Thesaurus

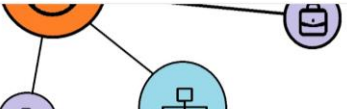
[Learn more](#)



NUTS Tree

NUTS Tree (Nomenclature of Territorial Units for Statistics)

[Learn more](#)



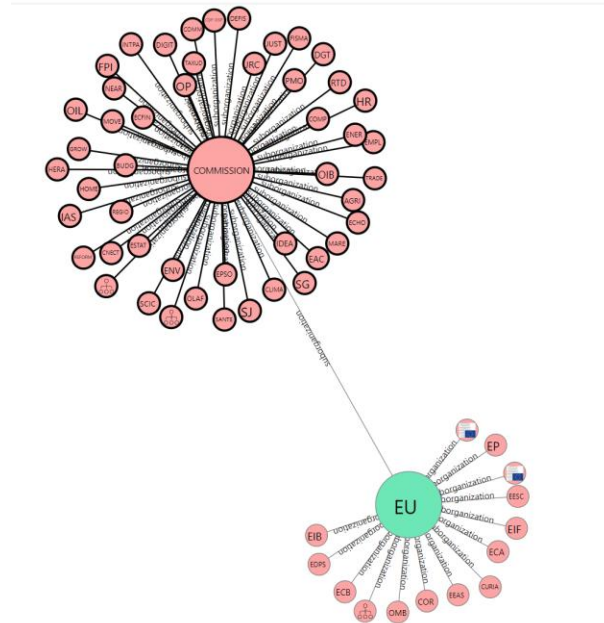
Authority table classification

Authority tables that link to other collections

[Learn more](#)

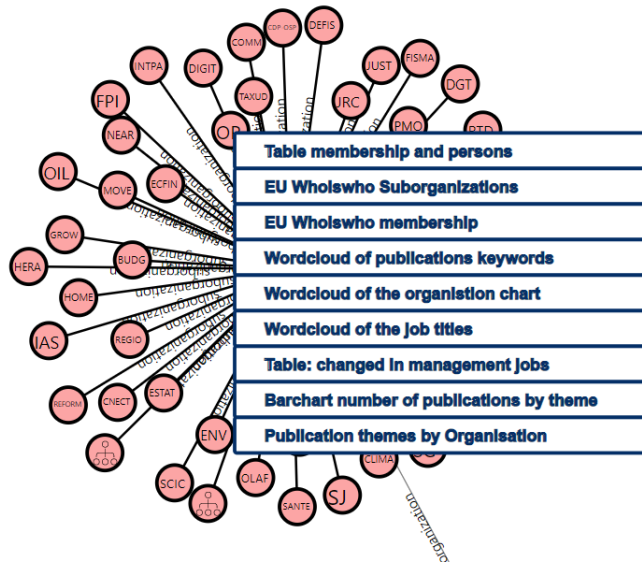
Visualisation and navigations of LOD graphs

- Demonstration: DGs of the Commission



Visualisation and navigations of LOD graphs

- Demonstration: visualisations options about “Publications Office”



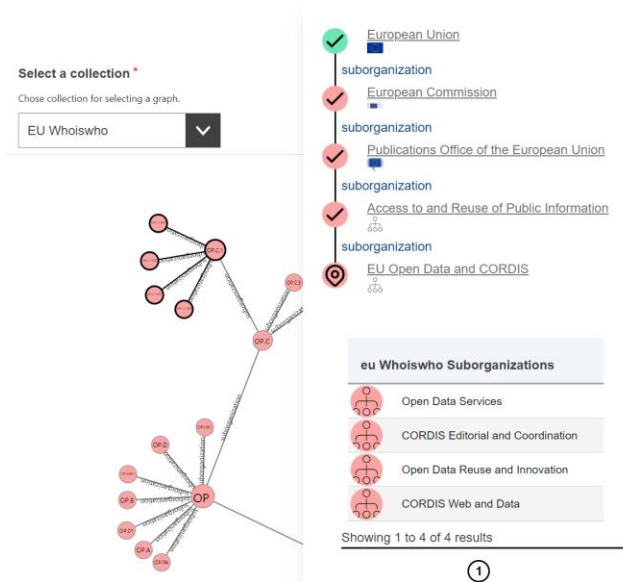
Visualisation and navigations of LOD graphs

- Demonstration: “word cloud” about “Publications Office” publications keywords



Visualisation and navigations of LOD graphs

- Demonstration: The open data team in the EU institutions graph



Visualisation and navigations of LOD graphs

- Demonstration: The list of MEPs in the AGRI committee of the EP

TABLE MEMBERSHIP AND PERSONS - COMMITTEE ON AGRICULTURE AND RURAL DEVELOPMENT			
MEMBERSHIP	PERSON	PERSON IMAGE	CITIZENSHIP
Chair	Norbert LINS		Germany
Vice-Chair	Mazaly AGUILAR		Spain
Vice-Chair	Benoit BITEAU		France
Vice-Chair	Daniel BUDA		Romania
Vice-Chair	Elsi KATAINEN		Finland

Visualisation and navigations of LOD graphs

- Demonstration: Wikipedia page of the chair MEP of the AGRI committee of the EP

The diagram on the left illustrates a network structure with a central red node and numerous peripheral red nodes, all interconnected by arrows, representing a dense web of relationships or data points.

The screenshot on the right shows the Wikipedia page for Norbert Lins, a German politician and Member of the European Parliament (MEP). The page includes a header with the Wikipedia logo, a navigation bar with tabs for 'Article' and 'Talk', and a search bar. The main content area features a portrait of Norbert Lins, his title as MEP, and a list of his political positions, including his role as chair of the Committee on Agriculture and Rural Development. The page also includes a 'Contents' section with links to various sections like 'Education and profession', 'Party', 'Political career', 'Political positions', 'References', and 'External links'. The 'Education and profession' section provides a detailed biography of Lins, mentioning his education at the University of Applied Sciences in Kehl and his career in the German military and public administration. The page footer includes a 'Print/export' section with links to download the page as PDF or print it, and a 'Languages' section with links to the page in German and French.

Visualisation and navigations of LOD graphs

- Demonstration: Catalogues of open data in data.europa.eu

Select a collection *

Chose collection for selecting a graph.

EU Data

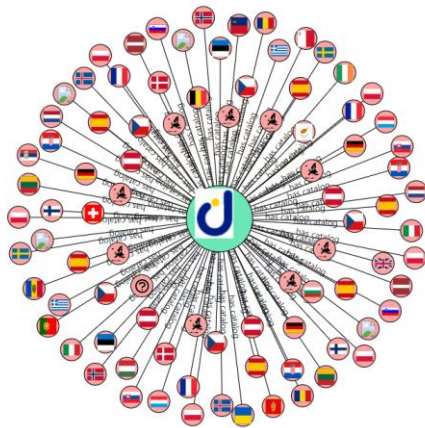
▼

Change graph

Show different graph in same collection.

Catalogues of datasets of

▼



Visualisation and navigations of LOD graphs

- Demonstration: Datasets of Spatial Data Portal Italy (in yellow it is grouped by themes). Dataset description and link to data.europa.eu

The screenshot displays the Vinalod interface, which is used for visualizing and navigating Linked Open Data (LOD) graphs. The interface includes a header with the European Commission logo and the Vinalod title. Below the header, there are tabs for Mode, Settings, Import/Export, Legend, and Manual and Live Demo. The Legend section shows a list of datasets, including 'data service', 'Catalog of open datasets', and 'Dataset documentation'. The Filter by section allows users to filter by 'Catalog of ...' and 'Data theme'. The main area shows a LOD graph with nodes and edges. A navigation panel on the right lists the datasets and their relationships, including 'data.europa.eu', 'Italian Catalogue of metadata for Spatial Data - Spatial Data Portal Italy', 'classified by', '04 POLITICS', 'has dataset', and 'Main manifolds in operation'. The panel also includes a table titled 'Main manifolds in operation' with columns for Description, Website link, and Quality. A large black arrow points from the navigation panel to the right, indicating a transition to the data.europa.eu portal.

European Commission

Vinalod

Mode Settings Import/Export Legend Manual and Live Demo

Legend

data service Catalog of open datasets Data theme

Dataset documentation

Filter by

+ Catalog of ...

+ Data theme

Apply

Navigation panel

data.europa.eu

has catalog

Italian Catalogue of metadata for Spatial Data - Spatial Data Portal Italy

classified by

04 POLITICS

has dataset

Main manifolds in operation

Main manifolds in operation

Description:	Website link:	Quality:
Main manifolds in operation	http://data.europa.eu/88u/dataset/infrastructures-sewerage-maincollectors	

data.europa.eu - The official portal for European data

Home Datasets Documentation Publications data.europa academy News & events Contact us

Home > Datasets > Main manifolds in operation

Dataset Main manifolds in operation

Italian Catalogue of metadata for Spatial Data

Dataset Quality Similar datasets

Main manifolds in operation

Dataset feed Linked data Cite

Languages Italian

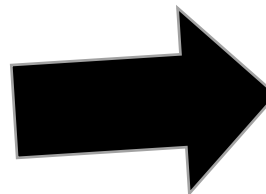
Contact E-Mail: tutela.acque@provincia.bz.it

Points URL: <https://ambiente.provincia.bz.it/acqua.asp>

Visualisation and navigations of LOD graphs

- Demonstration: Publication themes classification

The screenshot displays the Vinalod web application interface. At the top, the European Commission logo is visible. The main header is blue with the text "Vinalod". Below the header, there are tabs for "Mode", "Settings", "Import/Export", "Legend", and "Manual and Live Demo". The "Mode" tab is selected, showing options for "Chose mode" (Basic or Expert) and "Select a collection" (EU Publications). A "Filter by" section on the left allows filtering by "Directory O..." and "Corporate Body Type" (Delegation). The main area shows a graph visualization with a central node and many peripheral nodes. On the right, a sidebar titled "Publication Themes Classification" lists a hierarchy of nodes: "type", "Trade — Competition", "narrow", "External trade", "author", "Delegation to Canada", and "publish". Below this, a "Children" section shows "Manifestation by Expression" with a "pdf" result. At the bottom, it says "Showing 1 to 1 of 1 results".



Visualisation and navigations of LOD graphs

- Demonstration: Summaries of EU legislation classification

The screenshot displays the Vinalod application interface. At the top, a blue header bar contains the text "Vinalod". Below this, a navigation bar includes links for "Mode", "Settings", "Import/Export", "Legend", and "Manual and Live Demo". The "Mode" section is active, showing "Chose mode" with radio buttons for "Basic" (selected) and "Expert", and a note "Showing graphs in basic or expert mode". To the right, the "Select a collection" dropdown is set to "EU Law".

The main area features a graph visualization with nodes and edges. The nodes are colored in shades of blue, green, and red, and are connected by lines representing relationships. The graph is a complex network of interconnected nodes, with some nodes having multiple connections.

On the right side, a "Navigation panel" is visible. It contains a list of items with checkboxes and icons. The items are:

- ☒ Summaries of EU legislation classification
- ☐ type
- ☒ External trade
- ☐ narrower
- ☒ Development and sustainability
- ☐ classifies summary
- ☒ EU aid to non-euro area countries with trade deficit problems

Below the navigation panel, a section titled "EU aid to non-euro area countries with trade defi" provides metadata:

Created by agent:	http://publications.europa.eu/resource/body/PUBL
Date:	2016-01-27
Link:	https://eur-lex.europa.eu/legal-content/uri=legissum%3A25005

Visualisation and navigations of LOD graphs

- Demonstration: Expert mode starting navigation from URI

Vinalod

ModeSettingsImport/ExportManual and Live Demo

Mode

Chose mode *

Showing graphs in basic or expert mode

☐ Basic ☒ Expert

Start navigation from URI

Enter URI *

Select subject/object position *

▼

Show

Visualisation and navigations of LOD graphs

- Demonstration: Graph from URI:
<http://publications.europa.eu/ontology/cdm#institution> as Subject

The screenshot displays the Vinalod web application interface. At the top left is the European Commission logo. Below it, the 'Vinalod' header is visible. A navigation menu includes 'Mode', 'Settings', 'Import/Export', 'Legend', and 'Manual'. The 'Mode' section shows 'Chose mode' with options for 'Basic' and 'Expert' (selected). Below this is a 'Filter by' section with a search bar containing 'http://publi...' and buttons for 'Apply' and 'Clear'. A small graph visualization is shown next to the filter section. On the right, a 'Navigation panel' is open, displaying the URI 'http://publications.europa.eu/ontology/cdm#institution'. Below the URI, there are tabs for 'Children' and 'Details'. The 'Children' tab is active, showing a list of 6 results. Each result is represented by a colored circle and a text label. The results are as follows:

URI	Label
http://www.w3.org/1999/02/22-rdf-syntax-ns#type	http://www.w3.org/2002/07/owl#Class
http://www.w3.org/2002/07/owl#equivalentClass	nodeID:/b139287662
http://www.w3.org/2000/01/rdf-schema#label	Institution
http://www.w3.org/2000/01/rdf-schema#comment	Institution (typically public). The class institution has as instances institutions like the European Commission, the European Parliament, the European Council, the Council of the Regions etc. Institutions are defined by the corporate-body table
http://www.w3.org/2000/01/rdf-schema#subClassOf	http://publications.europa.eu/ontology/cdm#agent
http://www.w3.org/2000/01/rdf-schema#subClassOf	http://www.w3.org/2004/02/iskos/core#Concept

Showing 1 to 6 of 6 results

Visualisation and navigations of LOD graphs

- Demonstration: Options for URI <http://www.w3.org/2002/07/owl#Class>

Mode

Chose mode*

Showing graphs in basic or expert mode

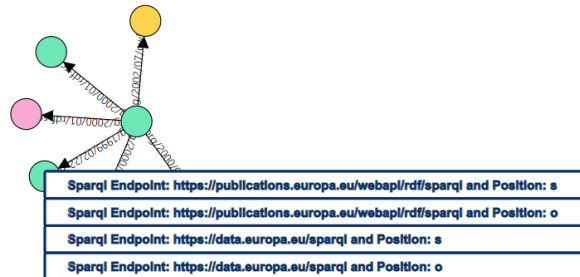
☐ Basic ☒ Expert

Filter by

+ [http://publi...](http://publications.europa.eu/webapi/rdf/sparql)

Apply

Clear



Visualisation and navigations of LOD graphs

- Demonstration: Results for URI <http://www.w3.org/2002/07/owl#Class> as Object

The screenshot displays the Vinalod web application interface. The top navigation bar includes 'Mode', 'Settings', 'Import/Export', 'Legend', and 'Manual and Live Demo'. The 'Legend' panel on the left lists various graph elements: 'uri' (green), 'link basic graph' (red), 'corporateBody' (purple), 'bnode' (yellow), 'typed-literal' (pink), and 'Publication Document' (grey). Below the legend is a 'Filter by' section with a list of URIs. The central area shows a LOD graph with nodes of different colors and sizes connected by lines. On the right, the 'Navigation panel' is open, showing a list of URIs with corresponding icons. Below this, the 'Children' section is visible, displaying a message about collapsed results and a list of URIs. At the bottom right, a pagination indicator shows 'Showing 1 to 2 of 2 results' and a page number '1'.

Vinalod

Mode Settings Import/Export Legend Manual and Live Demo

Legend

- uri
- link basic graph
- corporateBody
- bnode
- typed-literal
- Publication Document

Filter by

- + <http://publi...>
- + <http://www...>
- + <http://publi...>
- + <http://publi...>
- + <http://publi...>

Navigation panel

- <http://publications.europa.eu/ontology/cdm#institution>
- <http://www.w3.org/1999/02/22-rdf-syntax-ns#type>
- SPARQL Endpoint: <https://publications.europa.eu/webapi/rdf/sparql>
- <http://www.w3.org/2002/07/owl#Class>

Children Details

Several options displayed in graph. Click on each option to see results:

- [851 collapsed results](#)
- <http://www.w3.org/1999/02/22-rdf-syntax-ns#type>
- http://publications.europa.eu/ontology/cdm#agreement_non-member-states

Showing 1 to 2 of 2 results

1

Visualisation and navigations of LOD graphs

- Demonstration: Results for URI of an agreement with non-member states. Position: Subject

The screenshot shows the Vinalod interface. At the top left is the European Commission logo. Below it is a blue header with 'Vinalod' and a menu with 'Mode', 'Settings', and 'Import/Export'. To the right of the header is a 'Legend' section with three items: 'uri' (green circle), 'link basic graph' (red circle), and 'corporateBody' (purple circle). Below the legend is a 'Filter by' section with a search bar. On the right side, there is a 'Navigation panel' with a list of URIs and their corresponding icons (green checkmark, red X, or purple circle). The URIs are: <http://publications.europa.eu/ontology/cdm#institution>, <http://www.w3.org/1999/02/22-rdf-syntax-ns#type>, [SPARQL Endpoint: https://publications.europa.eu/webapi/rdf/sparql](https://publications.europa.eu/webapi/rdf/sparql), <http://www.w3.org/2002/07/owl#Class>, <http://www.w3.org/1999/02/22-rdf-syntax-ns#type>, [SPARQL Endpoint: https://publications.europa.eu/webapi/rdf/sparql](https://publications.europa.eu/webapi/rdf/sparql), http://publications.europa.eu/ontology/cdm#agreement_non-member-states, <http://www.w3.org/1999/02/22-rdf-syntax-ns#type>, [SPARQL Endpoint: https://publications.europa.eu/webapi/rdf/sparql](https://publications.europa.eu/webapi/rdf/sparql), and <http://publications.europa.eu/resource/cellar/6d8a3918-894a-4c75-ad6b-385316dc3a71>.

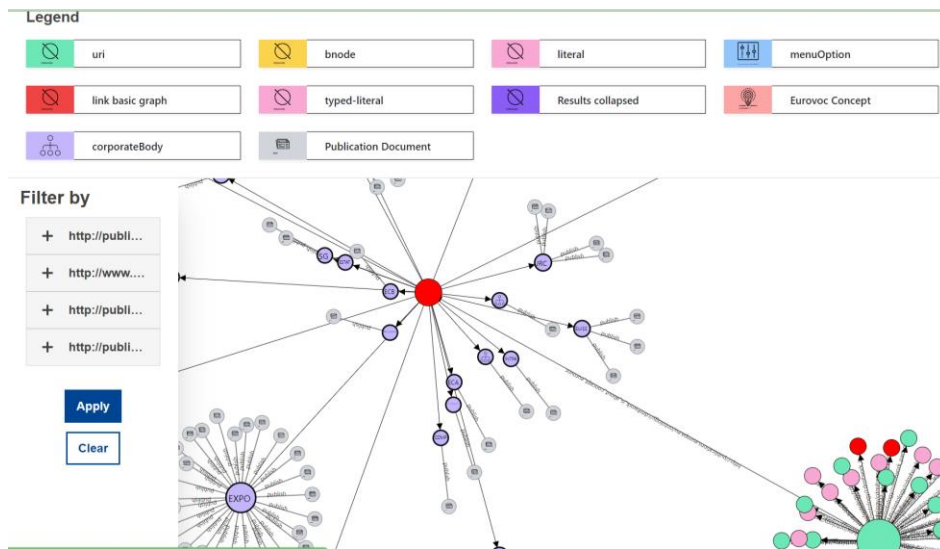
02/22-rdf-syntax-ns#type		http://publications.europa.eu/ontology/cdm#agreement_international
02/22-rdf-syntax-ns#type		http://publications.europa.eu/ontology/cdm#agreement_non-member-states
02/22-rdf-syntax-ns#type		http://publications.europa.eu/ontology/cdm#resource_legal
02/22-rdf-syntax-ns#type		http://publications.europa.eu/ontology/cdm#work
07/owl#sameAs		http://publications.europa.eu/resource/celex/21975A1223%2801%29
07/owl#sameAs		http://publications.europa.eu/resource/oj/JOL_1975_329_R_0001_006
a.eu/ontology/annotation#build_info		cdm:CDM_2.1.7 tdm:1523 xsdt:3945 saxon:9.0.0.1J JVM:1.6.0_29 metaconvJar:1.2.0 builddate:24/07/2014 16:16:40
a.eu/ontology/cdm#date_creation_legacy		1975-12-23
a.eu/ontology/cdm#resource_legal_based_on_concept_treaty		http://publications.europa.eu/resource/authority/treaty/TEEC_1957
a.eu/ontology/cdm#resource_legal_based_on_resource_legal		http://publications.europa.eu/resource/cellar/8204c9eb-d7d2-443d-a4c1-e6a72089338

Showing 1 to 10 of 50 results

1 2 3 4 5 Next >

Visualisation and navigations of LOD graphs

- Demonstration: Linking expert and basic graphs: Publications by Eurovoc concept Australia



Visualisation and navigations of LOD graphs

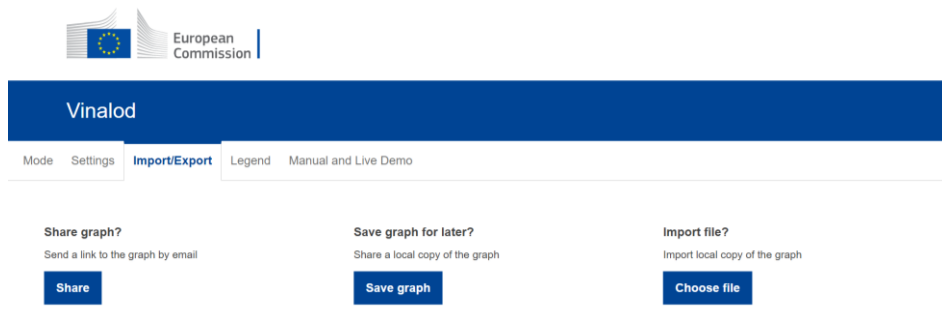
- Demonstration: Navigation panel for pdf from Eurovoc concept Australia

The image shows a web interface for navigating LOD graphs. On the left, a 'Navigation panel' lists several URIs with green checkmarks and circular icons indicating navigation status. The URIs include ontology classes like 'institution', 'type', 'Class', 'agreement_non-member-states', and 'work_is_about_concept_eurovoc', as well as a specific resource 'http://eurovoc.europa.eu/4160'. On the right, a detailed view of the selected resource is shown. It includes a vertical navigation bar with icons for 'author', 'publish', and 'manifestation type'. The 'author' is 'Directorate-General for Trade', and the 'publish' action is associated with the title 'Trade sustainability impact assessment in support of FTA negotiations between the European Union and Australia : final report.'. Below this, a 'pdf' icon is shown. At the bottom, a table provides metadata for the PDF:

pdf	
Creation date:	3/2/2023
lastModification:	3/2/2023
Link:	http://publications.europa.eu/a374-11ed-b508-01aa75ed7

Visualisation and navigations of LOD graphs

- Demonstration: Share or import graph



VINALOD: Visualisation and navigations of LOD graphs

- Next steps?
 - Extension of the tool
 - Integrate in data.europa.eu



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